Digital Twin Anchored Generation: Adaptive Questioning for Directed Movement in Opinion Space

Ishanu Chattopadhyay

***Abstract*—Can the choice of survey questions steer inferred ideology without controlling the answers? Digital Twin Anchored Generation (DTAG) uses Large Science Model (LSM) conditionals to select questions maximizing expected displacement along a pole-relative ideology index. In 960 GSS 2024 twenty-question simulations across six personas, directional policies shifted the index as intended without asking any of the 27 pole-defining items; 2,880 fixed-set replays preserved direction under reordering. In a prospective Prolific human study (34 completed sessions), mean shifts were -0.0783 (Left) and $+0.0382$ (Right), with 95% participant-bootstrap intervals excluding zero; Neutral and Random means were near zero. Right-policy respondents had both negative and positive initial indices. These results show that adaptive question choice can direct the ideological position inferred by a survey digital twin, making question-selection policy consequential for adaptive measurement.**

***Index Terms*—Predictive phenotypes, large science models, conditional response modeling, digital twins, survey methodology, opinion dynamics, adaptive questioning, generative simulation.**

I. Introduction

Survey questions are usually treated as measurements, but each answer also changes predictions about unanswered items. We ask whether choosing *which questions to pose* can steer a respondent's inferred ideological position. If so, adaptive surveys do more than collect information: their selection policies shape the modeled opinion profile.

Large Science Models (LSMs) provide a new framework for generative digital twins of multivariate systems. They learn interdependent conditional distributions, infer unobserved variables from partial evidence, and compare states through their predictive consequences. Related digital-twin applications include infant microbiome development [1] and influenza evolution through Emergenet [2]. DTAG applies an LSM to survey answers, defining a *predictive phenotype* over unasked items and its position relative to opposing reference opinion profiles. It selects questions to maximize expected signed displacement while respondents supply the answers.

We test this with 960 native GSS 2024 trajectories across six personas (excluding all pole-defining items), 2,880 fixed-set order replays, and 34 completed human sessions recruited through Prolific.

II. Related Work

Attitude dependencies need not align on one ideological axis [3], [4]. Partisan sorting and social influence shape these dependencies [5], [6]; cognitive consistency, wording, and order can affect reported attitudes [7]–[9]. This motivates separating information revealed by a question from any change induced by asking it.

Opinion-dynamics models often use low-dimensional transitions [10]–[12], while experiments on exposure and contact report heterogeneous effects [13]–[15]. Generative agents and prompted survey personas can produce plausible responses [16], [17]. DTAG instead combines learned high-dimensional survey conditionals with an explicit question-selection objective.

III. Large Science Models and the Predictive Phenotype

A. Conditional dependency model

Let $X = (X_1, \ldots, X_n)$ be categorical survey responses, with alphabets $\mathcal{A}_i$ and survey context c. An LSM comprises conditionals

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{1}$$

estimated by conditional inference trees [1], [18], [19]. Their dependencies are predictive, not causal; the conditionals need not form a coherent joint law [20].

A respondent state $s = \{X_j = a_j : j \in S\}$ holds observed responses. Unanchored variables remain missing in $\widetilde{x}_{-v}(s)$; each native tree averages over its missing branches. For unasked $v \notin S$,

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{2}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{3}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

B. Geometry as predictive equivalence

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{4}$$

Here D_{JS} denotes Jensen–Shannon divergence. The resulting predictive pseudometric distinguishes states by their conditional consequences.

Prior LSM reconstruction reduced mean distance distortion by 54.9% (GSS) and 60.9% (Eurobarometer) with 80% of responses masked [19].

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{5}$$

The reference poles specify opposing responses on 27 GSS 2024 items covering policy, civil liberties, spending, and religion (Table I) [19]. Other coordinates are unspecified; I_Φ measures predictive position rather than self-reported ideology.

Email: ishanu_ch@uky.edu Working research draft, October 2026. University of Kentucky. Native GSS 2024 computational experiments and human-response study.

TABLE I

GSS 2024 reference poles. Twenty-seven paired responses define s_L, s_R, with $I_\Phi = -1, +1$; all other coordinates are unspecified.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove

Item	Survey topic	Left	Right
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

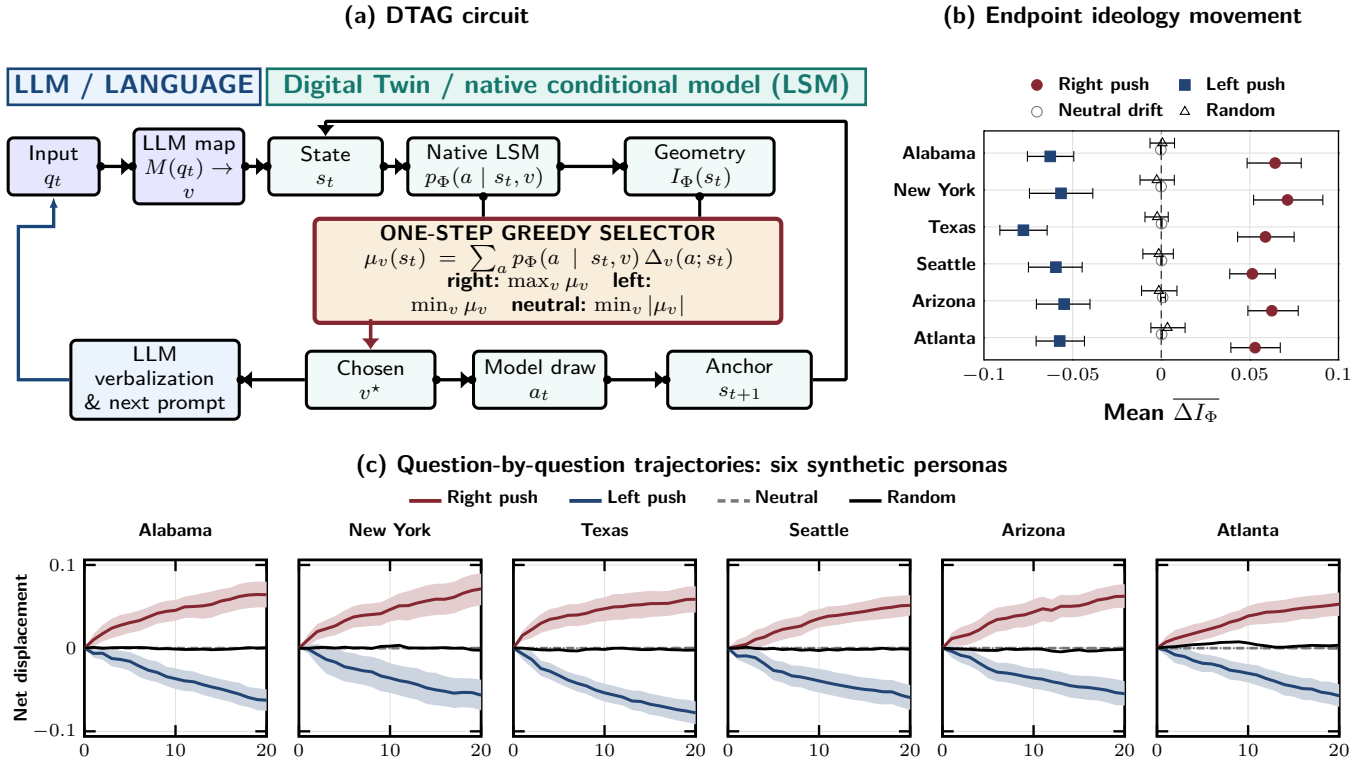


Fig. 1. **DTAG architecture and native adaptive results.** (a) LSM-anchored responses and directional question selection; language mapping is a separate interface. (b) Mean endpoint displacement across six personas with 95% bootstrap intervals. (c) Mean 20-question trajectories over 40 repetitions per persona and policy, with pointwise 95% intervals (right: red; left: blue; neutral: dashed gray; random: black).

IV. Adaptive Question Selection

A semantic interface can map a natural-language question q to a survey item $v \in M(q)$ and verbalize its answer. The native experiments use item identifiers directly without the LLM layer. For a selected question, DTAG samples a response and updates the predictive phenotype:

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \tag{6}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{7}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{8}$$

The initial persona anchors remain fixed. The optimizer chooses the question, not its answer (Fig. 1).

For candidate item v and answer a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{9}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v)\Delta_v(a; s), \tag{10}$$

$$\sigma_v^2(s) = \sum_a p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{11}$$

Right- and left-seeking policies maximize and minimize $\mu_v(s)$; neutral minimizes $|\mu_v(s)|$, and random samples an eligible item. By construction,

$$\mathbb{E}[I_\Phi(s_{t+1}) - I_\Phi(s_t) \mid s_t, v_t] = \mu_{v_t}(s_t).$$

The accumulated displacement depends on successive selected items and sampled answers.

The effect on an unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \quad (12)$$

A weighted sum $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$ measures redistribution across the remaining items. Unlike text retrieval [21], DTAG maintains an explicit conditional response state. The public implementation includes a web application (https://github.com/zeroknowledgediscovery/DTAG/tree/main/webapp).

V. Refinement and Possible Response Effects

Under *phenotypic refinement*, successive answers restrict predictions for unasked items. For held-out responses Y, the expected information from asking v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v|s}[H(Y \mid s, v = A_v)]. \quad (13)$$

Under a coherent joint model this is conditional mutual information, distinct from directional displacement $\mu_v(s)$. Held-out response prediction evaluates refinement directly.

A. Possible response effects.

Questioning a person may change later reports. A response–transition kernel separates an observed answer from a possible latent-state update:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \quad (14)$$

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; shifts in salience or belief can induce nontrivial transitions. A common held-out battery and delayed measurement can investigate these effects.

B. Order dependence.

For fixed realized answers, the terminal index is order-independent; sequentially generated answers need not be. A coherent joint distribution satisfies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \quad (15)$$

Separately fitted LSM conditionals may violate this identity [20]. Fixed-set replays evaluate its consequences for native response sequences.

VI. Native GSS 2024 Computational Experiments

A. Adaptive questioning from pinned persona states

We fixed six initial GSS 2024 persona states (Table II) and generated 40 independent response sequences per persona and policy (right, left, neutral, random). Each sequence comprised 20 questions with responses sampled from the native LSM; no LLM mapping or verbalization was used. *All 27 pole-defining items were excluded from selection in every policy* (Table I). Thus 960 adaptive trajectories contain 19,200 non-pole responses; repetitions share initial anchors but have independent sampled answers.

Across all six personas, right- and left-seeking policies moved the index in their intended directions (Fig. 1). As pole-defining questions were excluded, the movement results from conditioning on other survey items, not directly asking those used to define the reference index. Neutral and random policies provide comparisons.

B. Question selection as observed paths

Question identities are summarized in Table III. The 18 displayed heatmaps contain 161 distinct items across right, left, and neutral selection (Fig. 2). A shared categorical ordering approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \big|\pi(v_{purt}) - \pi(v_{pur't})\big|, \quad (16)$$

Here u indexes policies. Heatmap colors identify items, not ideological position; 67 further items occurred only in random selection.

C. Fixed-set replay: robustness to question ordering

For each persona and right/left/neutral policy, we froze the 20-item sequence from adaptive replicate zero, giving 18 fixed sets. Each set was replayed in original, reversed, and two shuffled orders with 40 fresh sampled responses per order (2,880 runs). Matched question-specific random variates reduced contrast variance.

The direction persisted across replays (Fig. 2): right-derived sets yielded mean displacement +0.0381 (original) and +0.0377 (reversed); left-derived sets yielded −0.0399 and −0.0410. In both orders, all six persona-specific 95% bootstrap intervals excluded zero in the prescribed direction. Shuffles followed the same pattern; neutral sets remained near zero.

All 54 pointwise bootstrap intervals for paired mean between-order contrasts included zero; no systematic mean order effect was detected. Selected question sets therefore retained directional displacement without continuous reoptimization.

In historical ten-question runs, baseline indices differed across environments despite matched persona, map, and pole hashes; native binaries were not compared. Their absolute indices are not combined with the present experiment.

VII. Human Study: Hypothesis and Validation

Our hypothesis is that adaptive questioning can *steer the inferred ideology index toward a specified pole*, with directional movement relative to the respondent's own baseline rather than requiring a particular starting ideology. Thus left-seeking questioning should give $\Delta I_\Phi = I_\Phi(s_{20}) - I_\Phi(s_0) < 0$, and right-seeking questioning should give $\Delta I_\Phi > 0$, across heterogeneous initial index values. Neutral and random policies provide non-directional comparisons.

We tested this in 34 completed human sessions with six common baseline questions and twenty adaptive questions (plus five attention checks): Left ($n = 5$), Right ($n = 17$), Neutral ($n = 7$), and Random ($n = 5$). Mean endpoint movements were −0.0783 for Left and +0.0382 for Right, with both 95% participant-bootstrap intervals excluding zero; Neutral (−0.0054) and Random (−0.0045) remained

TABLE II
PINNED GSS 2024 SYNTHETIC PERSONAS AND BASELINE POLE-RELATIVE INDEX.

Persona	**Pinned description**	**Initial $I_\Phi(s_0)$**
Alabama	45-year-old rural Alabama white man; married, two children; full-time farmer and veteran; high school; conservative, strong Republican; Protestant, weekly attendance; daily newspaper.	0.075706
New York	22-year-old New York City white woman; unmarried, no children; full-time retail worker, nonveteran; bachelor's degree; extremely liberal, strong Democrat; nonreligious; daily newspaper.	−0.163807
Texas	61-year-old small-town Texas white woman; married, three children; homemaker, high school; extremely conservative, strong Republican; Protestant, attends services several times weekly; daily newspaper.	0.045379
Seattle	34-year-old Seattle white man; unmarried, no children; employed full time, graduate degree; liberal, weak Democrat; nonreligious; newspaper several times weekly.	−0.119639
Arizona	39-year-old Mexican American man in an Arizona city; married, three children; employed full time, high school; conservative, weak Republican; Catholic, nearly weekly attendance.	−0.020437
Atlanta	47-year-old Atlanta Black woman; divorced, two children; employed full time, bachelor's degree; liberal, strong Democrat; Protestant, weekly attendance; daily newspaper.	−0.030807

Positive values lie nearer the right reference pole; negative values nearer the left.

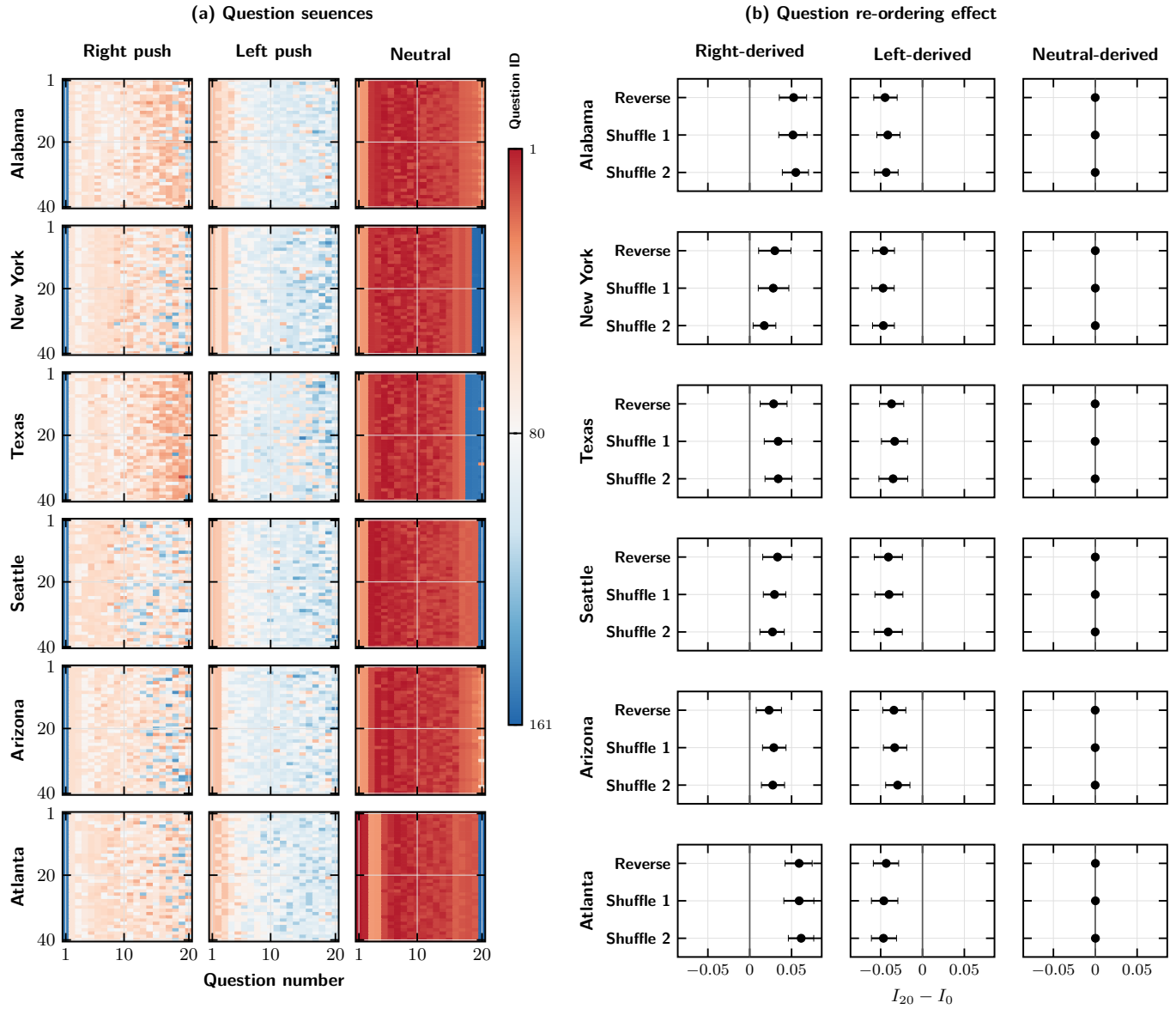


Fig. 2. **Selected questions and fixed-set replays.** Left: question-ID heatmaps for six personas and three selection policies (40 runs each); color denotes item identity. Right: mean index displacement and 95% bootstrap intervals for 20-item sets replayed in original, reversed, and two shuffled orders. Right- and left-derived sets retain their direction; neutral sets remain near zero.

TABLE III
Question selection in the native GSS 2024 experiment.

Selection policy	Distinct	Selections	Pole overlap
Right-seeking	100	4,800	0
Left-seeking	75	4,800	0
Neutral	43	4,800	0
Three-policy union	161	14,400	0
All four policies	228	19,200	0

The random policy contributed 4,800 selections; 67 of the 228 distinct items occurred only under random selection.

near zero. Right-policy respondents spanned both sides of the baseline ideology coordinate (11 negative, 6 positive), directly testing directional movement in a mixed-baseline cohort.

VIII. Human Response Study

A. Protocol and outcome

We conducted a prospective human-response study, recruiting participants through Prolific, an established online research-participation platform [22], to evaluate whether adaptive question-selection policies produce predictable displacement in the pole-relative LSM index. Participants provided informed consent before answering questions in the linked DTAG web application. Each participant completed six common baseline questions, up to twenty adaptive questions, and five interleaved attention checks. Participants responded in natural language; accepted survey-variable interpretations were incorporated into the conditional LSM state. The policy groups are *Left*, *Right*, *Neutral*, and *Random*. The original study randomized participants among all four policies. Subsequent cohorts used fixed-Left and fixed-Right policies. All results pool sessions by their assigned policy, regardless of recruitment wave.

Let s_0 denote the conditional state after the six baseline responses and s_t the state after t adaptive answers. The outcome at each step is the change in the fixed pole-relative geometric coordinate,

$$\Delta I_\Phi(t) = I_\Phi(s_t) - I_\Phi(s_0), \qquad t = 0, \ldots, 20. \tag{17}$$

A positive displacement is rightward in this model-defined coordinate and a negative displacement is leftward. The index measures the inferred survey-response phenotype and makes directional movement comparable across different baseline positions.

B. Participant characteristics and completion

Table IV reports the numbers of 20-question completers, the baseline index signs, and the mean question-20 displacement for each policy. The pooled data contain 49 completed sessions: 20 Left, 17 Right, 7 Neutral, and 5 Random. These comprise 24 completers from the original four-arm randomized study, 15 fixed-Left follow-up completers, and 10 fixed-Right follow-up completers. Endpoints are changes relative to each participant's own baseline; the later cohorts were not randomized against the earlier Neutral and Random groups.

C. Question-by-question trajectories and endpoints

Figure 3(a) plots mean $\Delta I_\Phi(t)$ for all four policies. For a given question number, the mean includes only participants with an observed answer at that step. A participant who did not finish contributes only through their last recorded answer, with no interpolation, imputation, or endpoint carry-forward. The shaded regions are pointwise 95% confidence intervals from participant-level bootstrap resampling (5,000 replicates by default). For clarity, *only Left and Right* carry shaded intervals; Neutral and Random are shown as mean trajectories.

Figure 3(b) reports the question-20 endpoint for *all four* policies, each as a mean and a 95% participant-bootstrap confidence interval. Unlike the stepwise trajectories, endpoint estimates use only participants with all twenty adaptive questions recorded. The zero reference is no change relative to baseline. The legend lists only the four policy names, and participant counts appear separately in Table IV.

D. Interpretation

The pooled human results support directional movement in the *model-inferred ideology index*. At question 20, Left ($n = 20$) had a mean change of -0.0993 (95% participant-bootstrap CI $[-0.1214, -0.0748]$); 19 of 20 moved left. Right ($n = 17$) had a mean change of $+0.0382$ (CI $[+0.0012, +0.0748]$); 11 of 17 moved right. Neutral ($n = 7$, -0.0054) and Random ($n = 5$, -0.0045) showed smaller mean changes. Left included seventeen negative and three positive baseline indices, and Right included eleven negative and six positive baselines.

Baseline-to-final correlations were high for both policies (Left: Pearson $r = 0.9518$; Right: $r = 0.9734$), consistent with substantial preservation of participant ordering under conditioning. The correlation between baseline ideology and displacement was $r = 0.2538$ ($p = 0.2803$) for Left and $r = 0.7314$ ($p = 0.00085$) for Right. These exploratory within-policy correlations do not by themselves establish a statistically significant difference in baseline dependence across policies.

The observed directional displacements concern conditional *inferred survey-response phenotypes*, not independently verified changes in participants' underlying political beliefs. Adding observations can refine the conditional phenotype without persuasion. Comparisons that combine the originally randomized arms with separately recruited fixed-policy follow-ups should be interpreted descriptively rather than as cross-wave randomized causal effects.

IX. Discussion and Conclusion

DTAG uses high-dimensional survey dependencies to select questions that systematically refine a predictive phenotype. The native experiments show directional movement using exclusively non-pole items; fixed-set replays

TABLE IV
HUMAN STUDY SUMMARY BY ADAPTIVE QUESTION-SELECTION POLICY*. ORIGINAL AND FOLLOW-UP SESSIONS ARE POOLED BY POLICY.

Policy	Completed n	Baseline $I_\Phi < 0$	Baseline $I_\Phi > 0$	Mean $\Delta I_\Phi(20)$
Left	20	17	3	−0.0993
Right	17	11	6	+0.0382
Neutral	7	6	1	−0.0054
Random	5	2	3	−0.0045

*Baseline signs refer to the inferred ideology coordinate before adaptive questioning. The final column is the mean observed 20-question change from baseline.

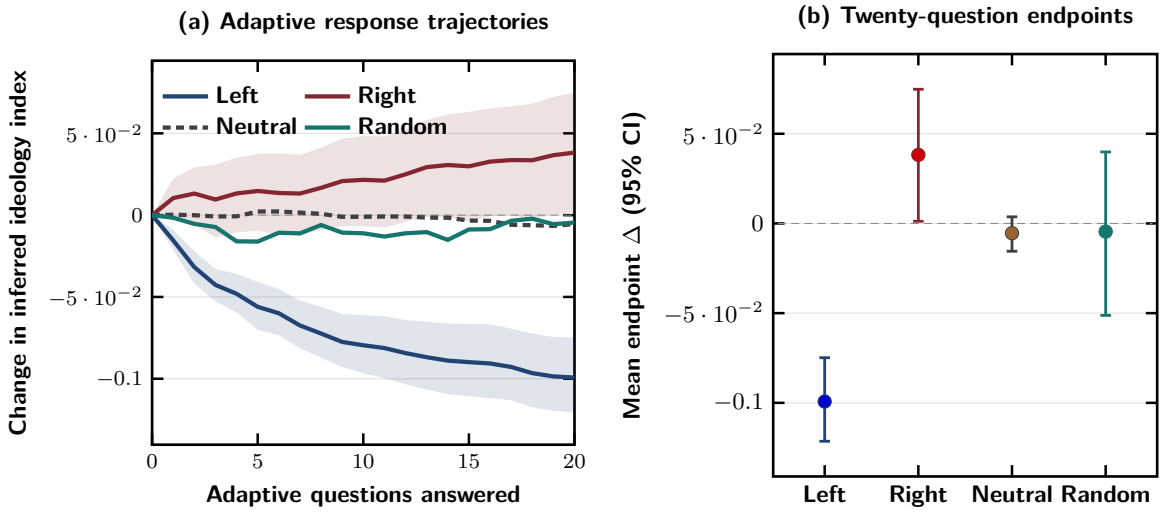


Fig. 3. **Human study response trajectories and endpoints.** (a) The mean change from baseline in the LSM-derived ideology coordinate as adaptive questions are answered, with pointwise 95% participant-bootstrap confidence intervals shaded only for Left and Right. Partial sessions contribute only through their last observed step. (b) Endpoint means and 95% bootstrap confidence intervals for all four policies, calculated from participants with all 20 adaptive answers. Arm sample sizes are reported in Table IV; follow-up and original sessions are pooled within Left and Right.

indicate that selected question identities, rather than their precise order, account for much of the directionality. The index depends on the reference poles, candidate pool, and conditional model and should be interpreted as predictive position rather than a self-reported ideology score.

The human-response study confirms that adaptive questioning can steer the inferred ideology index: left- and right-directed policies yielded signed mean shifts in the intended directions, while neutral and random policies remained near zero. These movements were measured relative to each participant's starting index, including a right-policy cohort with both negative and positive baseline values.

ACKNOWLEDGMENTS

This work was supported by the Defense Advanced Research Projects Agency (DARPA) under the MAGICS program (DARPA-EA-25-02-05-MAGICS-PA-025; Award No. HR0011-26-3-E016).

REFERENCES

[1] N. Sizemore, K. Oliphant, R. Zheng, C. R. Martin, E. C. Claud, and I. Chattopadhyay, "A digital twin of the infant microbiome to predict neurodevelopmental deficits," *Science Advances*, vol. 10, p. eadj0400, 2024.

[2] K. Wu, F. Li, and I. Chattopadhyay, "Emergenet: A digital twin of influenza a evolution for vaccine strain forecasting and emergence risk assessment," *Military Medicine*, p. usag255, 2026, published online 12 June 2026.

[3] P. E. Converse, "The nature of belief systems in mass publics," in *Ideology and Discontent*, D. E. Apter, Ed. New York: Free Press, 1964, pp. 206–261.

[4] J. R. Zaller, *The Nature and Origins of Mass Opinion*. Cambridge University Press, 1992.

[5] D. Baldassarri and A. Gelman, "Partisans without constraint: Political polarization and trends in american public opinion," *American Journal of Sociology*, vol. 114, no. 2, pp. 408–446, 2008.

[6] D. DellaPosta, Y. Shi, and M. Macy, "Why do liberals drink lattes?" *American Journal of Sociology*, vol. 120, no. 5, pp. 1473–1511, 2015.

[7] L. Festinger, *A Theory of Cognitive Dissonance*. Stanford: Stanford University Press, 1957.

[8] N. Schwarz, "Self-reports: How the questions shape the answers," *American Psychologist*, vol. 54, no. 2, pp. 93–105, 1999.

[9] C. S. Taber and M. Lodge, "Motivated skepticism in the evaluation of political beliefs," *American Journal of Political Science*, vol. 50, no. 3, pp. 755–769, 2006.

[10] M. H. DeGroot, "Reaching a consensus," *Journal of the American Statistical Association*, vol. 69, no. 345, pp. 118–121, 1974.

[11] N. E. Friedkin and E. C. Johnsen, "Social influence and opinions," *Journal of Mathematical Sociology*, vol. 15, no. 3–4, pp. 193–206, 1990.

[12] R. Hegselmann and U. Krause, "Opinion dynamics and bounded confidence models, analysis, and simulation," *Journal of Artificial Societies and Social Simulation*, vol. 5, no. 3, p. 2, 2002. [Online]. Available: https://www.jasss.org/5/3/2.html

[13] C. A. Bail, L. P. Argyle, T. W. Brown, J. P. Bumpus, H. Chen, M. B. F. Hunzaker, J. Lee, M. Mann, F. Merhout, and A. Volfovsky, "Exposure to opposing views on social media can increase political polarization," *Proceedings of the National Academy of Sciences*, vol. 115, no. 37, pp. 9216–9221, 2018.

[14] J. L. Kalla and D. E. Broockman, "The minimal persuasive

effects of campaign contact in general elections: Evidence from 49 field experiments," *American Political Science Review*, vol. 112, no. 1, pp. 148–166, 2018.
[15] D. E. Broockman and J. L. Kalla, "Durably reducing transphobia: A field experiment on door-to-door canvassing," *Science*, vol. 352, no. 6282, pp. 220–224, 2016.
[16] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate, "Out of one, many: Using language models to simulate human samples," *Political Analysis*, vol. 31, no. 3, pp. 337–351, 2023.
[17] J. S. Park, J. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein, "Generative agents: Interactive simulacra of human behavior," in *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.
[18] T. Hothorn, K. Hornik, and A. Zeileis, "Unbiased recursive partitioning: A conditional inference framework," *Journal of Computational and Graphical Statistics*, vol. 15, no. 3, pp. 651–674, 2006.
[19] H. Dahal, D. Young, J. Evans, and I. Chattopadhyay, "The fabric of opinion: How interdependent attitudes constrain world-views, predict behavior, and evolve over time," 2026, research manuscript, preliminary and not yet independently published at the time of this draft.
[20] D. Heckerman, D. M. Chickering, C. Meek, R. Rounthwaite, and C. Kadie, "Dependency networks for inference, collaborative filtering, and data visualization," *Journal of Machine Learning Research*, vol. 1, pp. 49–75, 2000. [Online]. Available: https://www.jmlr.org/papers/v1/heckerman00a.html
[21] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, "Retrieval-augmented generation for knowledge-intensive nlp tasks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474. [Online]. Available: https://arxiv.org/abs/2005.11401
[22] Prolific Research, "IRB applications and ethics approval," 2026, researcher Help Center, accessed October 2026. [Online]. Available: https://researcher-help.prolific.com/en/articles/449628-irb-applications-and-ethics-approval